

BIS Sahayak (Bureau of Indian Standards Assistant)

Comprehensive Project Context, Technical Architecture & User Manual • Smart India Hackathon 2026 (SIH26107)

Problem Statement: SIH26107 (Consumer & Industry Assistance)	Live Web Portal: https://sih2026-bis-assistant.vercel.app
Target Demographic: 1.4B Indian Citizens & MSME Manufacturers	Core Stack: Next.js 15, Gemini 2.5 Flash, pgvector RAG, React 19

1. Executive Summary & Problem Context

In India, the **Bureau of Indian Standards (BIS)** and the Ministry of Consumer Affairs govern thousands of official Indian Standards (IS) and mandatory **Quality Control Orders (QCOs)**. However, navigating complex compliance clauses, identifying counterfeit ISI marks, and understanding routine laboratory testing protocols is extremely difficult for both everyday consumers and MSME manufacturers.

BIS Sahayak bridges this gap by delivering an AI-driven, dual-persona, bilingual regulatory companion grounded in 21 authentic Indian Standards documents with zero-hallucination hybrid vector retrieval and Web Speech audio playback.

The Dual-Persona User Experience

Persona Mode	Target Demographic	Key Features & Behavior
Consumer Mode	Everyday shoppers, households, and general consumers	- Plain everyday language explanations - Counterfeit warning flags & fake mark alerts - What to look for when buying (e.g. pressure cookers, drinking water, toys) - Guidance on filing complaints on the official BIS Care Mobile App
Industry / MSME Mode	Manufacturers, QA engineers, testing labs, civil contractors	- Numerical acceptance thresholds & lab testing tolerances - Mandatory Quality Control Orders (QCO) gazette notifications - Routine vs. type testing procedures - Applicable Harmonized System (HS) trade classification codes

2. Key Platform Capabilities

- ■ Hybrid RAG Knowledge Pipeline: Grounds answers on 21 pre-indexed authentic Indian Standards (Steel, Electrical, Drinking Water, Appliances, Footwear, Cement, Toys, Batteries). Uses 768-dimensional normalized vectors with cosine similarity and domain-aware token weighting.
- ■ 100% Bilingual AI & UI (English + Hindi): Fully localized UI, prompts, citation chips, speech synthesis, and AI response generation in both natural Devanagari Hindi and English.
- ■ 4-Step ISI Mark Authenticity Verifier (Iverify): Validates authentic 7/8-digit CM/L license format under BIS Scheme-I and provides step-by-step verification guidelines for the BIS Care Mobile App.
- ■ Interactive Clause Drawer & Citations: Every response includes clickable citation chips that trigger a slide-out drawer or full detail page with clauses, limits, and testing parameters without leaving the chat.
- ■ Web Speech API Voice Playback: Native browser audio synthesis allowing users to listen to AI explanations in natural Hindi or Indian-accented English.

3. Step-by-Step User Manual (How to Use This Site)

- Step 1: Access the Application: Open <https://sih2026-bis-assistant.vercel.app> in any web browser on desktop or mobile.
- Step 2: Select Mode & Language: Use the top navbar to select **Consumer Mode** (for plain safety guidance) or **Industry Mode** (for technical parameters). Click the globe button to toggle between **English** and ■ ■ ■ ■ ■ ■ ■ ■.
- Step 3: Ask a Question or Choose a Sample Prompt: Type your question in the chat input (e.g. 'Is my pressure cooker safe?' or 'What are the tensile requirements under IS 1786 for Fe 500D?') or click any sample prompt pill in the hero banner.

- **Step 4: Read Structured Answers & Expand Technical Details:** The AI presents a concise 2-sentence direct answer followed by key bullet takeaways. Click the *'More Information & Technical Specifications'* accordion to expand detailed clause breakdowns.
- **Step 5: Listen with Voice Assist:** Click the speaker icon (🔊) next to any AI message to have the response read aloud.
- **Step 6: Inspect Standard Clauses:** Click on any **Cited Standard Pill** (e.g. *IS 2347:2017*) to slide out the interactive clause drawer and inspect testing parameters and conformity schemes.
- **Step 7: Verify ISI Mark & License (/verify):** Click **Verify ISI Mark** in the header. Enter a 7 or 8-digit CM/L license number (e.g. *CM/L-8400123*) to validate the license format and view BIS Care verification instructions.
- **Step 8: Browse Standards Catalog (/standards):** Navigate to **Standards Directory** to browse all 21 Indian Standards, filter by sector, category, or mandatory QCO status.

4. Technical Architecture & Ingestion Pipeline

Component	Technology	Role & Architecture
Frontend	Next.js 15 (App Router) React 19, TypeScript Tailwind CSS v4	Server and client component architecture, zero-flicker GovTech design system, responsive drawer inspection.
AI / Grounding	Google Gemini 2.5 Flash (@google/genai)	Multi-turn conversational RAG, temperature 0.3, bilingual prompt engineering, conversational fallback NLP.
Vector Retrieval	Neon Serverless PostgreSQL pgvector & In-Memory Store	768-dimensional normalized embeddings, hybrid cosine similarity + token weighting, zero-latency offline retrieval.
Quality Assurance	Vitest 4.1	Automated test suite verifying RAG retrieval, citation generation, bilingual translation, and embedding math (10/10 passing).

5. Pre-Indexed Indian Standards in Registry

- **IS 2347:2017:** Domestic Pressure Cookers (Burst pressure & safety valve release limits)
- **IS 14543:2024 / IS 10500:2012:** Packaged Drinking Water & Potable Drinking Water (Pesticide limits & sterility)
- **IS 1786:2008:** High Strength Deformed Steel Bars Fe 500D (Yield stress & elongation for seismic resistance)
- **IS 694:2010 / IS 1293:2019:** PVC Insulated Wiring Cables & Plugs / Sockets (Spark testing & flame retardancy)
- **IS 3196:2013:** Welded Low Carbon Steel LPG Cylinders (Hydrostatic burst testing)
- **IS 9873:2019:** Safety of Toys - Mechanical & Physical Properties (Small parts choking & flammability)
- **IS 16018:2021:** Protective Helmets for Two-Wheeler Riders (Impact absorption & retention systems)
- **IS 15298:2016:** Safety Footwear (Impact resistance 200J steel toecaps)

6. Official Government Portals & App References

Portal / Service	Official URL	Description
Bureau of Indian Standards	https://www.bis.gov.in	National standards body of India portal
e-BIS & Manakonline	https://www.services.bis.gov.in	Online license registry & standard sales
BIS Care Mobile App	Google Play Store / Apple App Store	Verify CM/L licenses & lodge consumer grievances
Ministry of Consumer Affairs	https://consumeraffairs.nic.in	Consumer protection & QCO orders